

Reviewer Pack — Minimal Deligne-Connes Tensor Rank and Communication Complex...

Entry 26dff10f6b65 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Minimal Deligne-Connes Tensor Rank and Communication Complexity Inequality

Entry ID: 26dff10f6b65 **Verdict:** INCONCLUSIVE **Recorded:** 2026-06-07 04:20:34 UTC

1. Conjecture

Field A (mathematical branch): Algebraic Geometry (Deligne-Connes Theory) **Field B** (complexity object): Communication Complexity

Statement:

For any given Boolean function f with n inputs, the minimal Deligne-Connes tensor rank ($\text{mDCT}(f)$) of its associated C-star algebra is linearly correlated with its communication complexity, such that $\text{mDCT}(f) = \Theta(C(f))$ where $C(f)$ is the communication complexity of f .

Rationale (proposer's reasoning):

The Deligne-Connes theory provides a noncommutative geometric framework for studying algebras arising from quantum systems. If this conjecture holds, it suggests a new approach to understanding the structure of Boolean functions and their communication complexity by leveraging the algebraic structures provided by Deligne-Connes theory.

Taxonomy category: DELIGNE_CONNES_TO_COMMUNICATION_COMPLEXITY (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 6ccc64484e409e9b

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

The conjecture is supported if the mean value of the ratio of minimal Deligne-Connes tensor rank ($\text{mDCT}(f)$) to communication complexity ($C(f)$) across 30 seeds is within a threshold of $\pm 10\%$ from the expected linear relationship, i.e., $|(mDCT_mean/C_mean) - 1| \leq 0.1$.

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	UNCERTAIN	1.00	UNCERTAIN	HITS
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - "Deligne-Connes tensor rank" AND "communication complexity" - "mDCT(f)" AND C(f) - "algebraic geometry" AND Deligne-Connes theory AND communication complexity

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.4s

5.1 Generated Python source

```
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def communication_complexity(f):
        n = int(math.log2(len(f)))
        count = 0
        for i in range(2**n):
            if f[i] != f[~i]:
                count += 1
        return count

    def minimal_deligne_connes_tensor_rank(f):
        n = int(math.log2(len(f)))
        rank = 0
        for i in range(n):
            for j in range(i+1, n):
                if f[i] != f[j]:
                    rank += 1
        return rank

    instances_tested = 0
    mDCT_sum = 0
    C_sum = 0
    n_max = 5

    for _ in range(30):
        n = random.randint(5, 40)
        if n > n_max:
```

```

        n_max = n

        f = generate_boolean_function(n)
        mDCT = minimal_deligne_connes_tensor_rank(f)
        C = communication_complexity(f)

        mDCT_sum += mDCT
        C_sum += C
        instances_tested += 1

    mDCT_mean = mDCT_sum / instances_tested
    C_mean = C_sum / instances_tested
    ratio = mDCT_mean / C_mean if C_mean != 0 else float('inf')

    conjecture_holds = abs(ratio - 1) <= 0.1

    return {
        "metric_name": "mDCT/C_ratio",
        "metric_value": ratio,
        "instances_tested": instances_tested,
        "n_max": n_max,
        "conjecture_holds": conjecture_holds,
        "counterexample": "" if conjecture_holds else "mapping_undefined"
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or list(range(2, 30*31, 2))

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    mDCT_mean = sum(r["metric_value"] for r in results) / len(results)
    C_mean = sum(1/r["metric_value"] for r in results if r["metric_value"] != 0) / len(results)
    support_fraction = sum(r["conjecture_holds"] for r in results) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mDCT_mean} std=NA support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mDCT_mean} std=NA support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"mapping_undefined\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

The test timed out before producing data, which means that the mean value of the ratio of minimal Deligne-Connes tensor rank to communication complexity could not be calculated. | next: Retry the experiment with a longer time limit or increase the computational resources to ensure the test completes and produces results.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	13582
2	preregistration	ollama_remote	glm4:latest	0	0	10683
3	novelty	ollama_remote	glm4:latest	0	0	8330
4	novelty	ollama_remote	glm4:latest	0	0	8805
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14120
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8876
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	16413
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8770
9	judge	ollama_remote	glm4:latest	0	0	21634

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 111215 ms total latency. Provider mix: {'ollama_remote': 9}

(full prompt+response transcripts available in research/audit/26dff10f6b65.jsonl)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in research/audit/26dff10f6b65.jsonl for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: research/pvsnp_sandbox/test_*.py (per-cycle)

Replay tarball: research/replay/26dff10f6b65.tar.gz (if generated)